

GREEN-FINANCE: REAL-TIME CARBON & ENERGY OBSERVABILITY FOR FINANCIAL MICROSERVICES

Abstract :

In the era of rapid digital transformation, financial institutions are scaling their cloud infrastructure without a clear understanding of their digital carbon footprint. Current monitoring solutions focus on performance and cost but lack granular visibility into energy consumption at the microservice level. This project proposes a "Green-Finance" observability framework designed to track the real-time electricity usage and carbon intensity of individual banking services, such as payment processing and authentication.

Leveraging a modern tech stack, the system utilizes FastAPI for a distributed microservices architecture, Kepler (eBPF-based metrics) for kernel-level energy telemetry, and Prometheus for data aggregation.

The results are visualized through custom Grafana dashboards, providing a "Carbon-per-Transaction" metric.

This project bridges core engineering concepts from Operating Systems and Computer Networks with modern DevOps practices. By identifying energy-inefficient code "hotspots," the framework enables organizations to reduce operational costs and fulfill environmental sustainability goals (ESG compliance), aligning with the global shift toward green computing.

I. Introduction

Introduce the problem or opportunity the project addresses. Include background information, significance, and relevance to current technology or research.

II. Problem Definition / Statement

Background

Rapid advancements in Artificial Intelligence (AI), the digital economy, and data-driven technologies have accelerated economic growth but also introduced environmental challenges, social inequalities, and governance concerns. At the same time, global efforts are shifting toward sustainable and circular economic models, supported by ESG principles and AI-enabled education.

Problem Statement

The research addresses the challenge of balancing digital and economic growth with environmental sustainability and social equity. While emerging technologies create new opportunities, they also increase carbon emissions, resource consumption, and ethical concerns related to data governance and accessibility.

Significance of the Study

The study focuses on the **Triple Bottom Line (TBL)** by promoting:

- **Environmental sustainability** through green finance and reduced carbon emissions.
- **Social equity** through circular economy practices and inclusive development.
- **Responsible digital governance** by protecting data sovereignty and ethical AI use.

Relevance to Current Technology and Research

Real-time Energy Monitoring for Financial Microservices project aligns with current

advancements in sustainable finance and digital technologies:

- Green Finance: Financial institutions are increasingly adopting green finance practices to promote sustainable investments, reduce carbon emissions, and support environmentally responsible decision-making.
- ESG-Based Decision Making: Environmental, Social, and Governance (ESG) metrics have become key indicators for evaluating organizational sustainability, encouraging businesses to integrate carbon footprint and energy efficiency into their operations.
- Real-Time Monitoring: Modern monitoring tools and dashboards enable continuous tracking of energy consumption and operational performance, providing organizations with actionable insights for reducing their digital carbon footprint.
- Sustainable Financial Technology: The growing adoption of digital financial services and green banking has increased the need for energy-efficient, transparent, and sustainable financial systems. Integrating real-time energy monitoring with financial microservices supports this objective by improving visibility, accountability, and informed decision-making.

Current research highlights:

- Sustainable AI using energy-efficient Small Language Models (SLMs).
- AI readiness assessment through validated measurement scales.
- Open innovation and AI competencies for entrepreneurship.
- Leveraged advanced analytics for sustainability-focused decision-making.

III. Objectives

- Develop a system to analyze and reduce CO₂ emissions through green banking practices.
- Implement a standardized framework for sustainable finance and green bond evaluation.
- Incorporate ESG scores and carbon footprint metrics into sustainable investment decision-making.

IV. Literature Survey / Review

Green Finance and Environmental Sustainability

The study highlights the role of green finance in promoting sustainable economic growth and reducing carbon dioxide emissions, particularly in BRICS economies. It emphasizes that green financial initiatives encourage environmentally responsible investments and support climate risk mitigation. The paper also recommends that financial institutions actively promote green finance to achieve long-term sustainability. These findings motivate the integration of energy monitoring and sustainability metrics into financial systems.

Investor-Driven Financial Performance

This paper discusses the evolving role of investors as active participants in shaping corporate financial performance. It explains how Environmental, Social, and Governance (ESG) metrics have become important factors in investment decisions, encouraging organizations to improve sustainability practices alongside profitability. The study also stresses the importance of standardized ESG reporting and carbon emission disclosures, which align with the objectives of the proposed monitoring framework.

Global Investment Trends in the Digital Economy

The World Investment Report 2025 examines recent trends in sustainable finance and digital infrastructure investments. It reports significant growth in sustainable finance products and green bonds while highlighting the increasing investments in digital technologies such as data centers and semiconductor industries. The report emphasizes the need for energy-efficient digital infrastructure and standardized sustainability reporting, providing strong support for developing real-time energy monitoring solutions for financial microservices.

Research Gap

Existing studies primarily focus on green finance policies, ESG reporting, and sustainable investment trends. However, limited attention has been given to real-time monitoring of energy consumption at the financial microservice level. Current approaches lack integrated systems that combine energy monitoring, ESG analytics, and carbon footprint visualization within financial applications. The proposed project addresses this gap by developing a

real-time monitoring dashboard integrated with a payment portal, enabling organizations to track energy usage, monitor sustainability metrics, and support green finance initiatives.

V. Methodology

Technologies and Tools

- **FastAPI** – Backend API development
- **Prometheus & Grafana** – Real-time energy monitoring and dashboards
- **Docker** – Microservices deployment
- **Python** – Data processing and analytics
- **SQLite/PostgreSQL** – Data storage

System Design and Architecture

- **Microservices Architecture** for modular financial services.
- **Real-time Monitoring Layer** to track energy usage per microservice.
- **ESG Analytics Module** for sustainability metrics.
- **Interactive Dashboard** for visualization and reporting.

Algorithms and Techniques

- **System Dynamics Modeling** for sustainability analysis.
- **Quantile Regression** for carbon emission impact analysis.
- **Real-time Data Analytics** for energy consumption monitoring.
- **Carbon Footprint Estimation** using energy usage metrics.

Implementation Plan (ADDIE Model)

1. **Analysis:** Identify energy-intensive financial microservices.
2. **Design:** Develop the system architecture and ESG monitoring framework.
3. **Development:** Build APIs, monitoring modules, and visualization dashboards.
4. **Implementation:** Deploy and test the monitoring system.
5. **Evaluation:** Assess energy savings, carbon reduction, and system performance.

VI. Project Scope

Included

- Real-time monitoring of energy consumption in financial microservices.
- Microservices-based architecture for modular banking services.
- ESG metrics and carbon footprint tracking.
- Interactive dashboards for energy usage visualization.
- Energy-efficient AI models (e.g., TinyLlama) for analytics.
- Sustainability reporting using standardized ESG frameworks.

Excluded

- Large Language Models (LLMs) with high computational requirements.
- High-power, energy-intensive hardware configurations.
- Traditional financial accounting and profitability analysis (e.g., ROA, ROE, Tobin's Q).
- Monitoring of private or non-financial organizations.
- Non-standardized or incomplete sustainability reporting.

VII. Expected Outcomes

Key Deliverables

- Real-time dashboard for monitoring energy consumption of financial microservices.
- ESG-based monitoring framework for sustainability tracking.
- Energy-efficient AI module using Small Language Models (SLMs).
- Interactive reports and visualizations for energy usage and carbon footprint.

Performance Metrics

- Reduced energy consumption of monitored microservices.
- Improved visibility into real-time energy usage and carbon emissions.
- Accurate monitoring and reporting of ESG-related metrics.
- Low-latency, scalable, and reliable monitoring system.

Impact

- Promotes sustainable software development practices.
- Supports green banking and ESG-driven decision-making.
- Enables organizations to monitor and reduce their digital carbon footprint.
- Raises awareness of energy-efficient AI and cloud computing practices.

VIII. Timeline

Phase	Duration	Activities
Phase 1: Planning & Environment Setup	Week 1	Requirements gathering, technology selection, database design, project setup, authentication, dashboard UI prototype.
Phase 2: Dashboard Development	Weeks 2–3	Develop the real-time energy monitoring dashboard, ESG metrics visualization, carbon footprint analytics, APIs, and backend integration.
Phase 3: Payment Portal Development	Weeks 4–5	Design and implement the payment website for carbon credit purchases, transaction management, payment gateway integration, and receipts.
Phase 4: Integration & Testing	Weeks 6–8	Integrate dashboard and payment portal, implement APIs, perform functional testing, security testing, and bug fixing.

Phase 5: Deployment & Monitoring	Weeks 9–10	Deploy the application using Docker, configure monitoring dashboards (Grafana/Prometheus), and optimize performance.
Phase 6: Documentation & Final Evaluation	Weeks 11–12	Prepare project documentation, user manual, presentation, demonstration, and final evaluation.

IX. Resources Required

Hardware Resources

- Laptop/Desktop (minimum Intel i5/Ryzen 5, 8–16 GB RAM)
- Stable Internet Connection
- Development workstation for coding and testing
- Cloud platform (optional: AWS, Azure, or Render) for deployment

Software Resources

- **Frontend:** HTML, CSS, JavaScript, React.js
- **Backend:** FastAPI (Python)
- **Database:** PostgreSQL
- **Monitoring:** Prometheus & Grafana
- **Containerization:** Docker
- **IDE:** Visual Studio Code
- **Version Control:** Git & GitHub
- **API Testing:** Postman
- **UI Design:** Figma (optional)

Datasets

- Sample financial transaction data
- Energy consumption logs of microservices
- ESG and carbon emission datasets (sample/public datasets)

- Carbon emission conversion factors

Human Resources

- **Frontend Developer** – Dashboard and Payment Portal UI
- **Backend Developer** – API development and business logic
- **Database Administrator** – Database design and management
- **Project Guide/Supervisor** – Technical guidance and review
- **Testers** – Functional and integration testing

XI. Conclusion

The Green-Finance: Real-time Energy Monitoring for Financial Microservices project provides an effective solution for monitoring the energy consumption and carbon footprint of digital financial services. By integrating a real-time monitoring dashboard with a secure payment portal, the system enables organizations to track sustainability metrics, visualize energy usage, and manage carbon credit transactions efficiently. The use of energy-efficient technologies and ESG-based analytics supports informed decision-making while promoting green banking practices. Overall, the project demonstrates how technology can contribute to environmentally sustainable and transparent financial systems.

References

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